Externalities

Reference: Chapter 10, Mankiw

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The idea

- **Consumption Externality.** Consumption that has an effect on others' production/consumption.
- **Production Externality.** Production that has an effect on others' production/consumption.
- The key here is these are goods that people care about (for better or worse) that are not sold in the market, and hence, do not have an explicit price/cost.

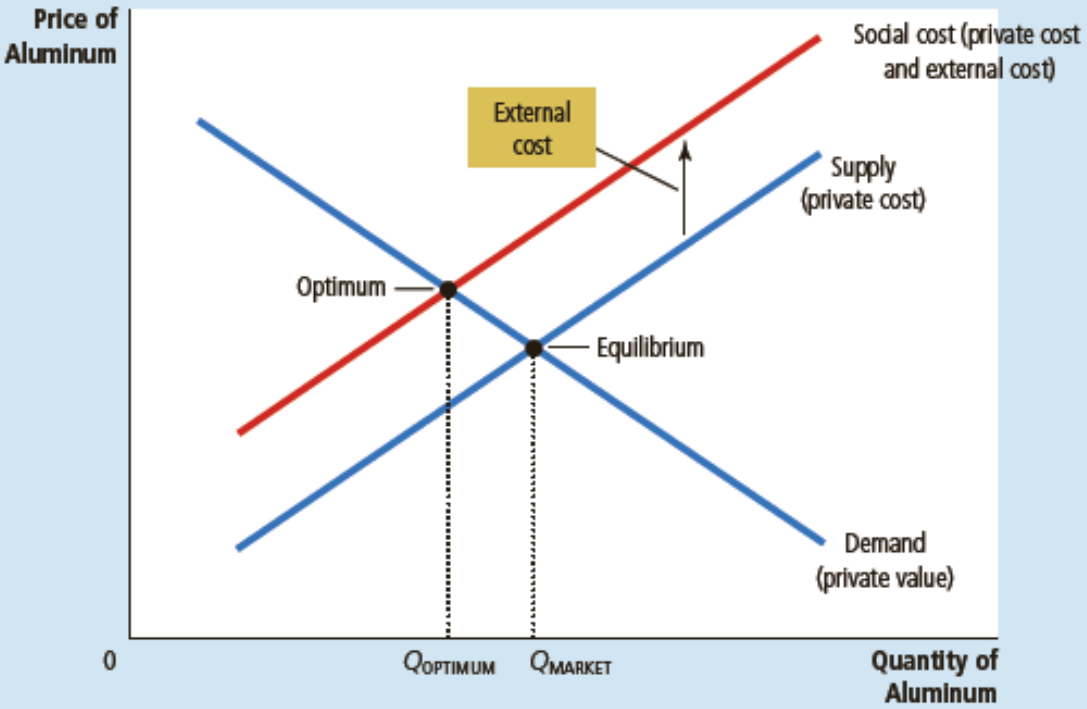
Smokers vs Non-Smokers

- Externalities can be studied in a more formal way, but requires a little more knowledge of economics.
- We will try to understand it in a broader sense, with some graphical interpretation.
- Then, for us, the most important thing becomes policy and its relation to externalities.
- How do we see this in the example mentioned in the title of the slide?

A basic graphical interpretation

FIGURE 2

Pollution and the Social Optimum
In the presence of a negative externality, such as pollution, the social cost of the good exceeds the private cost. The optimal quantity, $Q_{OPTIMUM}$, is therefore smaller than the equilibrium quantity, Q_{MARKET}.



How it looks

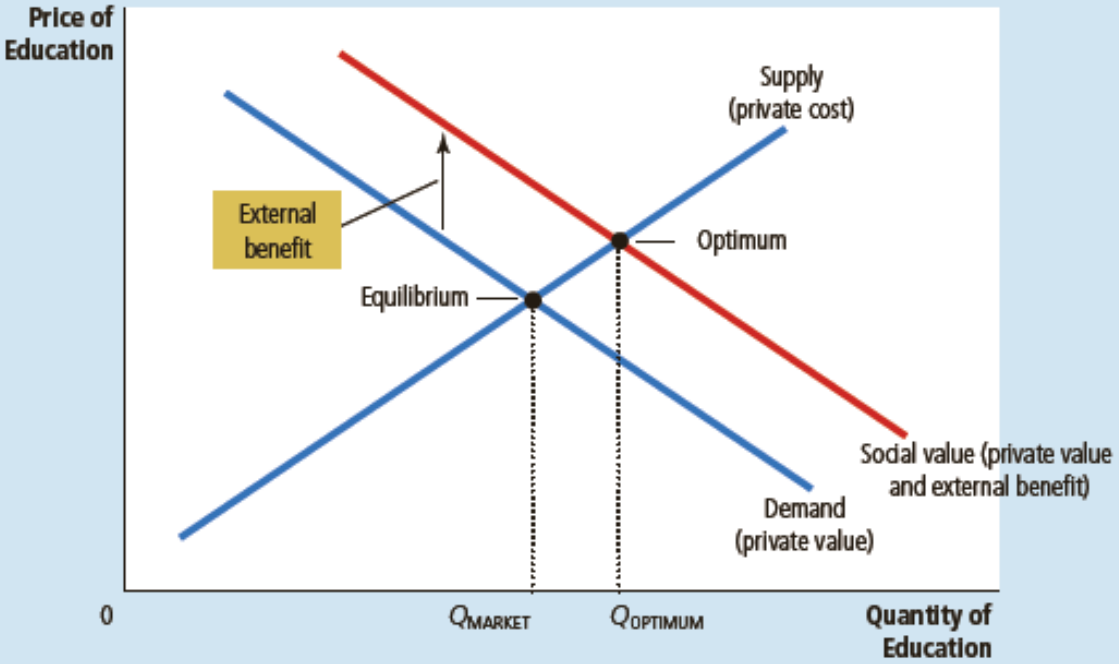
- This inefficiency occurs because the market equilibrium reflects only the private costs of production.
- How to obtain the social optimum?
- Tax. How?
- The use of such a tax is called **internalizing the externality**

What about positive?

FIGURE 3

Education and the Social Optimum

In the presence of a positive externality, the social value of the good exceeds the private value. The optimal quantity, $Q_{OPTIMUM}$, is therefore larger than the equilibrium quantity, Q_{MARKET}.



What happens here?

- Education definitely has private value.
- However, it also has positive externalities.
- This is true for both school and higher education.
- Subsidies.

Important Example

- **Technology Spillover.**
  - Good because it increases production.
  - Bad because it affects incentives.
  - Possible policy include subsidy and IPR.

Command-and-Control Policies

- One way is to simply forbid (outlaw) certain behaviour.
- Note that this approach is quite costly, as it generally requires monitoring.
- Additionally, it is hard to make pollution out and out illegal.
- This can work in specific cases, or at times in a seasonal manner (like GRAP policies in Delhi).
- However, this is not a comprehensive approach.

Taxes

- Taxes on negative externalities are called Pigovian taxes.
- What is better, prohibition (quota) or tax?
- Why are they good?
- Asymmetric effects and can make more sense.
- Promote innovation to reduce costs.
- In general, you will notice that quotas/prohibition are only useful in very specific settings.

Trading quotas

- Suppose everyone is allowed to pollute upto some quantity X.
- However, firm 1 may not need to pollute that much while firm 2 may need to pollute more.
- If they were allowed to trade, pollution would remain the same but firms would be better off.
- This is also a market mechanism. It internalizes the externality.

Coase Theorem

A famous result, called the **Coase theorem** after economist Ronald Coase, suggests that it can be very effective in some circumstances. According to the Coase theorem, **if private parties can bargain over the allocation of resources at no cost**, then the **private market will always solve the problem** of externalities and allocate resources efficiently.